

Recent announcements



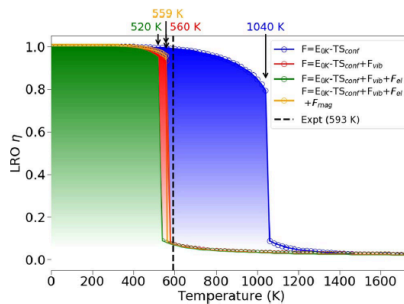
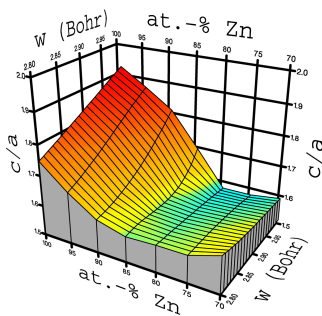
Updated Formula Collection

(https://canvas.kth.se/courses/62610/discussion_topics/551994)

Dear all, thank you for the seminar yesterday. I have now uploaded an updated formula collection, which is the same version that you will be provided during the exam. Note that I corrected a typo in the expression for the shear modulus in the Voigt bound, both in the formula collection and in the solution to Exercise 5. Best regards, Stephan

Posted on:

20 May 2026, 08:52



Hexagonal Ag-Zn alloys from Quantum Mechanical modeling

Ordering temperature in tetrataenite (FeNi) at various QM modeling levels

Welcome to the course MH1036!

We are glad to welcome you to MH1036. During this course, you will learn about the classical and quantum mechanical models of the intrinsic properties of materials. After a brief introduction to classical material theories, basic quantum mechanics will be applied to free electrons, and to electrons in metals and semiconductors. Describing materials using quantum mechanical equations assumes the knowledge of crystal lattices and reciprocal space, which is part of the course. Electronic structure methods will be briefly introduced. Theories of lattice vibration and elasticity will be presented, as well as electronic structures of common metals and semiconductors. A phenomenological approach to magnetism as well as the origin of magnetism will be discussed.

Remember that you must have registered to maintain access to your courses after the web registration period.

We hope that you, by the end of this course, will be able to explain basic physical concepts in crystal structures, electronic structures, lattice vibrations and magnetism. You will be able to analyze and solve problems for which atom level physics required. You will also gain a fundamental basis to describe, explain and predict properties of solid materials from atom level.

Course content

In the Syllabus, you can find a course summary and links to the Course PM and the course's intended learning outcomes (ILOs).

Intended learning outcomes

After passing the course, the student should be able to:

1. Explain basic physical concepts in crystal structures, electronic structures, lattice vibrations and magnetism.
2. Analyze and solve problems where the atom level approach is to prefer over the macroscopic (or technical) approach.

For higher grades, the student should furthermore be able to:

3. Describe, explain and predict properties of solid materials from atom level.

Lectures and exercises

#	Week	day/month	time	room	Activity	person
1	W12	17/3	10-12 am	U21	Introduction Lecture 1: Classical theory of metals	Levente Vitos
2	W12	20/3	10-12 am	U21	Lecture 2: Electromagnetic waves and materials	Levente Vitos
3	W12	20/3	1-3 pm	U21	Exercise 1	Stephan Schönecker
4	W13	24/3	1-3 pm	U21	Lecture 3: Fundamentals of quantum mechanics	Levente Vitos
5	W13	27/3	10-12 am	U21	Lecture 4: Schrödinger equation	Levente Vitos

6	W13	27/3	1-3 pm	U21	Exercise 2	Stephan Schönecker
7	W14	31/3	10-12 am	Q33	Lecture 5: Free electrons in metals	Levente Vitos
8	W14	1/4	1-3 pm	U21	Lecture 6: Structures of solids	Levente Vitos
9	W16	2/4	10-12 am	U21	Exercise 3	Stephan Schönecker
10	W16	16/4	1-3 pm	U21	Lecture 7: Electrons in periodic potentials	Levente Vitos
11	W16	17/4	1-3 pm	U21	Lecture 8: Band structure	Levente Vitos
12	W17	21/4	10-12 am	U21	Lecture 9: Electronic structure methods	Levente Vitos
13	W17	21/4	1-3 pm	U21	Exercise 4	Stephan Schönecker
14	W18	29/4	1-3 pm	U21	Lecture 10: Cohesive properties	Levente Vitos
15	W18	30/4	10-12 am	U51	Lecture 11: Atomistic theory of elasticity	Levente Vitos
16	W19	5/5	10-12 am	U31	Lecture 12: Lattice vibrations	Levente Vitos
17	W19	5/5	1-3 pm	U31	Exercise 5	Stephan Schönecker

18	W20	7/5	10-12 am	U51	Lecture 13: Magnetism	Levente Vitos
19	W20	8/5	10-12 am	W38	Lecture 14: Semiconductors	Levente Vitos
20	W21	11/5	10-12 am	U31	Exercise 6	Stephan Schönecker
21	W21	19/5	8-12 am	U31	Seminar	Levente Vitos, Stephan Schönecker
22	W22	26/5	8 am - 1 pm	Q13, Q15, Q17, Q24	Written examination/TEN1	Levente Vitos, Stephan Schönecker

Information and assignments

You find published information about the course, and published course content, in Modules. You also find the Zoom link for our scheduled online activities in the module.

Under Assignments, you will find all quizzes and submissions that are due during the course. Note that you might find assignments in the modules; it's the same assignment, just linked to from the module.

Course literature

James D Livingston: **Electronic Properties of Engineering Materials**

(Lecture 1: Chapter 1; Lecture 2: Chapter 2; Lecture 3: Chapter 8; Lecture 4: Chapter 9; Lecture 5: Chapter 12; Lecture 7: Chapters 11 and 12; Lecture 8: Chapter 13; Lecture 12: Chapter 7; Lecture 14: Chapter 15)

N. Ashcroft, N. Mermin: **Solid State Physics**

(Lecture 6: Chapters 4 and 7; Lecture 10: Chapter 20; Lecture 12: Chapter 22 and 23; Lecture 13: Chapter 33; Lecture 14: Chapter 28 and 29)

G. Grimvall: **Thermophysical Properties of Materials**

(Lecture 10: Chapter 3; Lecture 11: Chapter 4; Lecture 12: Chapter 6)

Russell C. Hibbeler: **Modern Physics, University Physics with Modern Physics**, EBook, Global Edition, 2015 (electronic access via KTH library)

(Lecture 3: Chapters 40 and 41)

Grades

You can find your grades on submitted assignments and quizzes under Grades.

Grading scale

A, B, C, D, E, FX, F

Examination

- INL1 - Homework, 1.5 credits, grading scale: P, F
- TEN1 - Written exam, 6.0 credits, grading scale: A, B, C, D, E, FX, F

Homework/INL1: homework should be submitted by 14th of April (first part) and 15th of May 2025 (second part).

Written examination/TEN1: 26th of May (W22), 8 am - 1 pm, rooms Q13, Q15, Q17, and Q24.

Communication and groups

You can find all announcements posted via Announcements.

Under the People tab, you can find the groups you are a part of in this course.

Course evaluation board

To improve the quality of the course, an opportunity is offered to create a course evaluation board (kursnämnd) where a few student representatives can easily interact with the course coordinator regarding issues related to the course structure.

The requirement for participating in the board is to be enrolled in the course. Students select the board representatives (1-3 people).

Once you have formed the board, you can schedule a meeting with us if needed. If we do not hear from the board, we will assume that everything is going smoothly.

Contact information

In this course, you will meet Prof. Levente Vitos, and Doc. Stephan Schönecker.

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